

Gaseous State

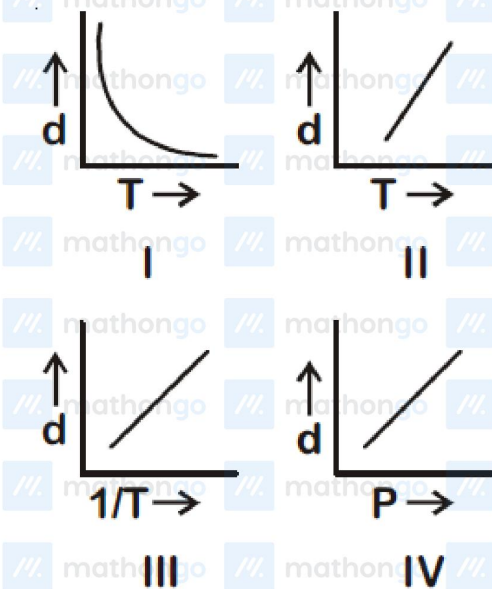
Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q1 JEE Main 2020 - 2 September (Morning)

Which one of the following graphs is not correct for ideal gas?



d = Density, P = Pressure, T = Temperature

- (A) II
- (B) III
- (C) I
- (D) IV

Q2 JEE Main 2020 - 3 September (Evening)

A mixture of one mole each of H_2 , He and O_2 each are enclosed in a cylinder of volume V at temperature T . If the partial pressure of H_2 is 2 atm, the total pressure of the gases in the cylinder is

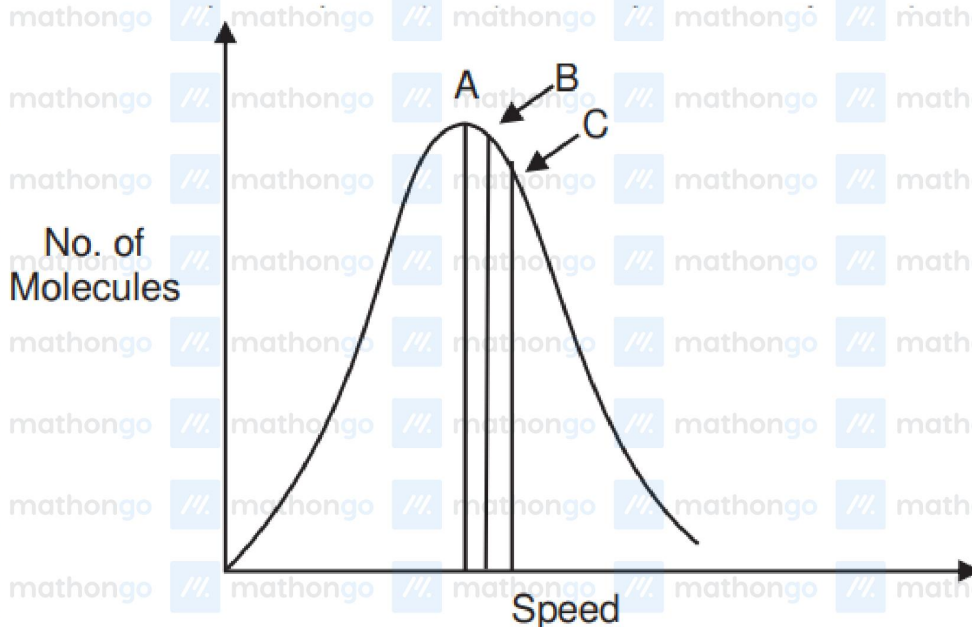
- (A) 14 atm
- (B) 38 atm
- (C) 22 atm
- (D) 6 atm

Q3 JEE Main 2020 - 6 September (Morning)

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A spherical balloon of radius 3cm containing helium gas has a pressure of 48×10^{-3} bar. At the same temperature, the pressure, of a spherical balloon of radius 12cm containing the same amount of gas will be $\times 10^{-6}$ bar.

Q4 JEE Main 2020 - 7 January (Evening)

- (A) $A = C_{MPS}$, $B = C_{Average}$, $C = C_{RMS}$
(B) $A = C_{Average}$, $B = C_{MPS}$, $C = C_{RMS}$
(C) $A = C_{RMS}$, $B = C_{Average}$, $C = C_{MPS}$
(D) $A = C_{MPS}$, $B = C_{RMS}$, $C = C_{Average}$

Q5 JEE Main 2020 - 9 January (Evening)

A cylinder containing an ideal gas (0.1 mol of 1.0 dm^3) is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of ethylene glycol at its freezing point. If the stoppers S_1 and S_2 (as shown in the Figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be

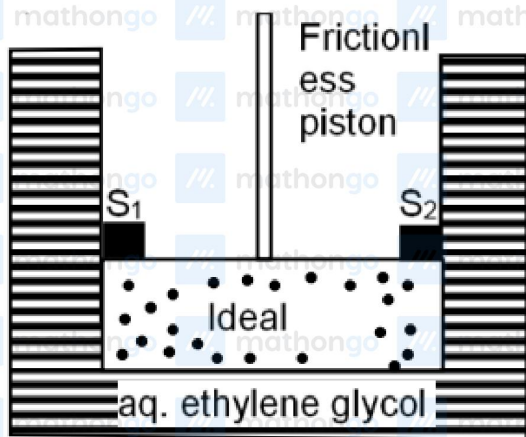
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Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

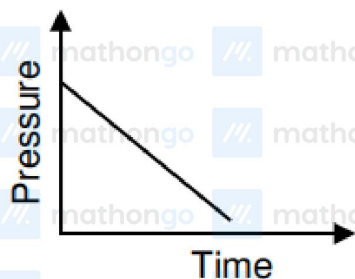
(Given, $K_f(\text{water}) = 2.0 \text{ K kg mol}^{-1}$, $R = 0.08 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$)



Q6 JEE Main 2020 - 9 January (Evening)

A mixture of gases O_2 , H_2 and CO are taken in a closed vessel containing charcoal. The graph that represents the correct behaviour of pressure with time is:

(A)



Gaseous State

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

(B) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo



(C) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo



(D) mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo mathongo



Gaseous State

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (A)

Q2 (D)

Q3 (750)

Q4 (A)

Q5 (2.18)

Q6 (B)

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Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q1

$$PM = dRT$$

$$\Rightarrow d \propto \frac{1}{T}$$

$$\Rightarrow d \propto P$$

Q2

Pressure due to 1 mole of H_2 gas = 2 atm

$\therefore$ Pressure due to 1mol of H_2 + 1mol of He + 1mol of O_2 = 6atm

Q3

At constant temperature and number of moles,

$$P_1 V_1 = P_2 V_2$$

$$P_1 = 48 \times 10^{-3} \text{ bar}; V_1 = \frac{4}{3}\pi(3)^3$$

$$V_2 = \frac{4}{3}\pi(12)^3$$

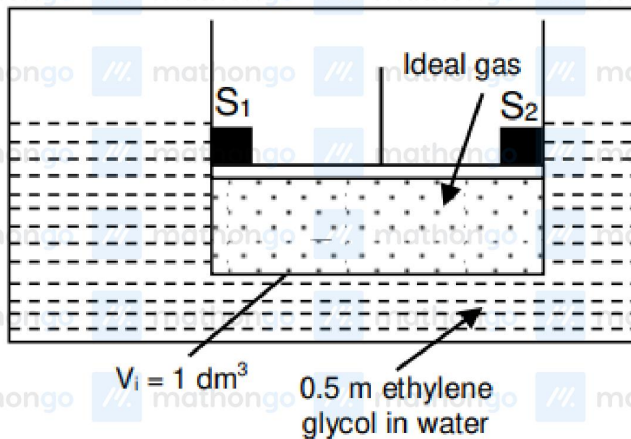
$$P_2 = \frac{P_1 V_1}{V_2} = \frac{48 \times 10^{-3} \times (3)^3}{(12)^3}$$

$$= \frac{48 \times 10^{-3}}{64} = 7.5 \times 10^{-4} = 750 \times 10^{-6} \text{ bar}$$

Q4

$$C_{RMS} > C_{Avg} > C_{MPS}$$

Q5



$$K_f = 2.0$$

$$m = 0.5 \text{ m}$$

$$\Delta T_f = K_f m$$

$$= 0.5 \times 2$$

$$T_{\text{initial}} = 272 \text{ K}$$

$$n = 0.1 \text{ mol}$$

$$V = 1 \text{ dm}^3$$

$$P_{\text{gas}} = \frac{nRT}{V} = \frac{0.1 \times 0.08 \times 272}{1}$$

$$= 2.176 \text{ atm}$$

After releasing piston $P_1 V_1 = P_2 V_2$

$$2.176 \times 1 = 1 \times V_2$$

$$V_2 = 2.176 \text{ dm}^3 \simeq 2.18 \text{ dm}^3$$

Q6

Theory based